MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, DIODE, SILICON TYPES
1N483B, 1N485B, 1N486B, 1N5194, 1N5195, 1N5196
JAN, JANTX, JANTXV, AND JANS

This amendment forms a part of Military Specification MIL-S-19500/118D dated 11 August 1986, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 2

Following 3.3.1, add the following paragraphs:

"3.3.2 Double plug construction. Double plug construction is one where the terminal plugs have equal nominal diameters and are in direct contact with the metallization materials of the semiconductor die; the use of a point contact whisker or other wire conductors is not allowed.

"3.3.3 Thermally matched. The coefficients of thermal expansion of die and package materials shall be matched to prevent thermal stress of dissimilar materials during thermal excursion.

"3.3.4 Noncavity. The package glass must be in intimate contact with the semiconductor die isolating the anode and cathode regions, and insuring immunity from particle related failures. Voids may be present provided isolation (to prevent arcing) and particle immunity are insured.

"3.3.5 1N5194, 1N5195, and 1N5196. These diodes shall be of metallurgically bonded, double plug construction utilizing high temperature metallurgical bonding between both sides of the silicon die and terminal pins.

"3.3.6 JANS construction. These devices shall be metallurgically bonded, thermally matched, noncavity, double plug construction, utilizing high temperature metallurgical bonding between both sides of the silicon die and terminal pins. Opaque glass is prohibited. The silver button or dumet package design is not permitted. No solder or low temperature bonding materials shall be used in the diode assembly."

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4.3.1, delete in its entirety and substitute the following:

"4.3.1 Power burn-in and steady state operation life. All devices shall be operated under one of the following conditions.

Type 1/	T_A = +25°C, ±3°C, I_O = 200 mA f = 60 Hz	T_A = +25°C ±3°C
1N483B, 1N5194	V_{RWM} = 70 V (pk)	I_F = 175 mA
1N485B, 1N5195	V_{RWM} = 180 V (pk)	I_F = 175 mA
1N486B, 1N5196	V_{RWM} = 225 V (pk)	I_F = 175 mA

1/ Derate at 1.2 mA/°C between +25°C and +150°C.
Derate at 1.0 mA/°C above +150°C."

PAGE 6

TABLE IIa., subgroup 5, conditions column, delete in its entirety and substitute "see 4.3.1".

PAGE 7

TABLE IIb., subgroup 3, conditions column, delete in its entirety and substitute "see 4.3.1".

PAGE 8

TABLE III, subgroup 6, conditions column, delete in its entirety and substitute "see 4.3.1".

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - MI
Navy - SH
Air Force - 11, 85, 99
DLA - ES

User activities:
Navy - AS, CG, MC, OS
Air Force - 14, 19

Preparing activity:
Navy - EC

Agent:
DLA - ES

(Project 5961-1120)

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